

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking, supporting distributed training and inference across multiple nodes.
- Engineered a pluggable distributed executor routing workloads across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation under scarce-capacity constraints.
- Built a deployment-gated evaluation framework enforcing guardrails and quality/regression checks across model outputs, blocking 15% of failed releases before traffic cutover.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended and optimized inference framework internals (vLLM-class serving) with kernel-level development (CUDA/Triton), distributed KV cache, and speculative decoding to improve LLM serving performance.
- Cut p99 inference latency 420ms to 140ms across 50K+ daily requests by migrating synchronous serving to async vLLM with autoscaling.
- Developed multi-modal computer vision models (PyTorch), running model training and fine-tuning on multi-GPU clusters with compression and quantization for efficient inference.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts across 25+ services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed storage engine in C++ handling concurrent writes with linearizable reads and log compaction; a from-scratch study of distributed systems internals.

LLaMA 2 Fine-Tuning Pipeline for Financial Documents

github.com/ganeshhgupta/llama2-finance-finetuning

- Post-trained LLaMA 2 with LoRA and PEFT, running systematic hyperparameter experiments on multi-GPU hardware to study capacity-performance tradeoffs.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production LLM eval and observability platform with latency tracking and regression detection across model versions; FastAPI backend, PostgreSQL.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Go, Rust, Java, SQL

Inference Engineering: vLLM, CUDA/Triton Kernel Development, Distributed KV Cache, Speculative Decoding, Quantization, Multi-GPU/Multi-Node Serving

ML: PyTorch, Model Training/Fine-Tuning, LoRA/PEFT, Computer Vision, Model Evaluation

Infrastructure: Multi-Cloud GPU Clusters (AWS, GCP, CoreWeave), InfiniBand/RoCE, Distributed Systems, Docker, Kubernetes

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer